

Outperforming atmospheric EHD propulsion performances in strong adverse flows

José M. D. C. Marques, David Fabre and Franck Plouraboué
IMFT, Toulouse

European
Innovation
Council



cnrs



iPROP

This project has received funding from the European Union's Horizon Europe Research and Innovation Programme under grant agreement No 101098900. Views and opinions expressed are however those of the authors(s) only and do not necessarily reflect those of the European Union or European Innovation Council and SMEs Executive Agency (EISMEA). Neither the European Union nor the granting authority can be held responsible for them.

Introduction to Electro-hydro-dynamics (EHD)

Electro-hydro-dynamic (EHD) systems rely on acceleration of a charged fluid, directly converting electrical power into mechanical power.

Chapman first modeled the behavior of intensity with an external flow :

$$I = CV(V - V_i) + Ku(V - V_i) \quad V_i : \text{inception voltage} \quad C, K : \text{constants}$$

Seville Chapman, Corona point current in wind, Journal of Geophysical Research, 75(12):2165-2169, April 1970.

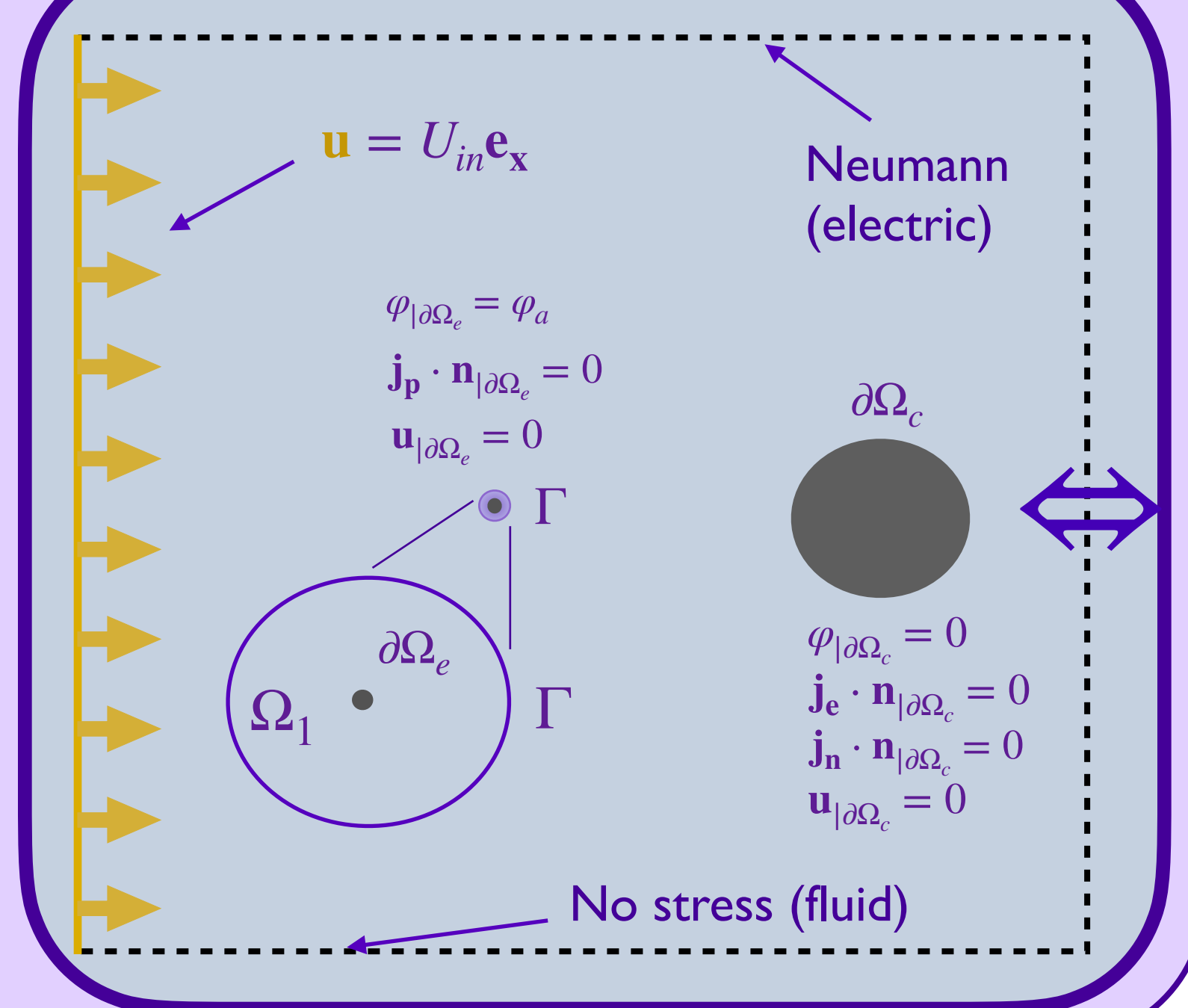
Free electrons in the atmosphere are accelerated against the electric field lines and collide against neutral particles. After the collisions electrons will be **ejected** from the neutral particles, thus **generating positive ions** by following the reaction : $A + e^- \rightarrow A^+ + 2e^-$. This process will repeat itself exponentially and lead to an **electronic avalanche** that creates a **cold plasma** and **ionizes** air, making it conductive.

Positive ions will be accelerated along the electric field lines and **collide** with **neutral particles** in air. Through **momentum transfer** they will generate the so called **"ionic wind"**.

Monrolin, Étude théorique et expérimentale de la propulsion électrohydrodynamique dans l'air. INPT, 2018.
Monrolin & Plouraboué, J. Comp. Physics, 443:110517, October 2021.

How can EHD be modeled in the presence of an external flow ?

Typical Configurations and Boundary Conditions



Dimensionless Parameters

Electro-drift Péclet : $P_e = \mu \phi_a / D_\rho$

EHD electric speed : $U_e = \frac{\phi_a}{r_c} \sqrt{\frac{\epsilon_0}{\rho_f}}$

Reference speed : $U_{ref} = U_{in}$

Reynolds : $R_e = U_{in} r_c / \nu_f$

Massuda : $M_a = \frac{U_e^2}{U_{in}^2}$

EHD Macro-to-Micro number : $M_\rho = \frac{U_{ref}}{\mu E}$

Townsend law : $\alpha = \beta_g e^{-\frac{E_i}{E}}$

Two-Way Dimensionless Corona Discharge and Fluid Model

Modeling of a fully coupled corona discharge with the external fluid flow can be achieved by adding a fluid speed to the convective part of the charge conservation equations and an electric forcing to the Navier-Stokes equations. The dimensionless two-way coupled model is solved in **two regions**, one near the corona where the cold plasma is set up, and the other on the drift region.

$$\begin{aligned} \nabla \cdot \mathbf{E} &= n_e + n_n - n_p \\ \nabla \cdot [n_p(\mathbf{E} + M_\rho \mathbf{u}) - 1/P_e \nabla n_p] &= \alpha ||n_e \mathbf{E}|| + S_{ph} \\ \nabla \cdot [n_e(-\mathbf{E} + M_\rho \mathbf{u}) - 1/P_e \nabla n_e] &= (\alpha - \eta) ||n_e \mathbf{E}|| + S_{ph} \\ \nabla \cdot [n_e(-\mathbf{E} + M_\rho \mathbf{u}) - 1/P_e \nabla n_n] &= \eta ||n_e \mathbf{E}|| \\ \mathbf{u} \cdot \nabla \mathbf{u} &= -\nabla p + \frac{\Delta \mathbf{u}}{R_e} - M_a(n_p - n_e - n_n)\mathbf{E} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

Linear asymptotic expansion with respect to the

EHD Macro-to-Microscopic speeds : $M_\rho \approx 10^{-2}$

$$\Rightarrow \mathbf{X} = \mathbf{X}^0 + M_\rho \mathbf{X}^1 + O(M_\rho^2)$$

Picella, Fabre & Plouraboué, AIAA J., p. 1-12, April 2024.
Marques, Fabre & Plouraboué 2025 J. Phys. D: Appl. Phys.

Low Speed Regime : $U_e > U_{in}$

$$U_{ref} = U_e \Rightarrow M_\rho = \frac{U_e}{\mu E} = \frac{1}{\mu_p} \sqrt{\frac{\epsilon_0}{\rho_f^*}} \approx 10^{-2} \quad \& \quad M_a = 1$$

M_ρ is the ratio of macroscopic to microscopic EHD speeds. This parameters is very small, and thus the coupling is only **one-way**, from the plasma onto the fluid.

High Speed Regime : $U_e < U_{in}$

$$U_{max} = U_{ref} \Rightarrow M_\rho = \frac{U_{in}}{\mu E} \quad \& \quad M_a = U_e^2 / U_{in}^2$$

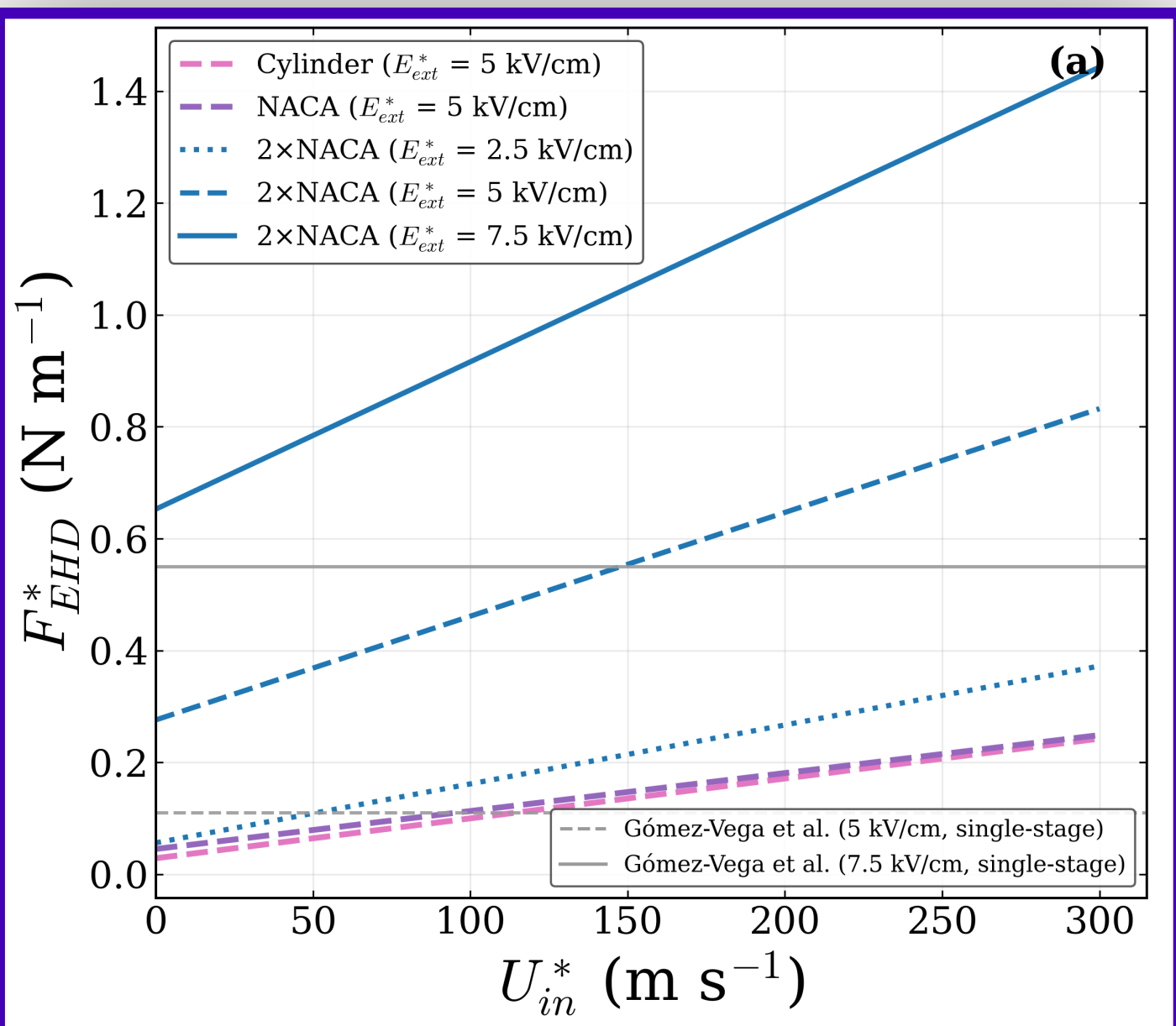
M_a decreases **quadratically** with U_{in} , whereas M_ρ increases. The one-way **reversed**, from the fluid onto the plasma !

Propulsive characteristics

Revised Thrust

$$F_{EHD} = \int_{\Omega} \rho \mathbf{E} \cdot \mathbf{u} \approx \frac{Id}{\mu} + \frac{\epsilon_0}{\mu} \int_{\Omega} \mathbf{E} \cdot \nabla \mathbf{v}$$

Generated EHD Thrust



- EHD thrust more than **doubles** with adverse flow
- Up to **7 times** the reference value from literature

N. Gomez-Vega & S. R. H. Barrett, AIAA Journal 2024 62:4, 1342-1353

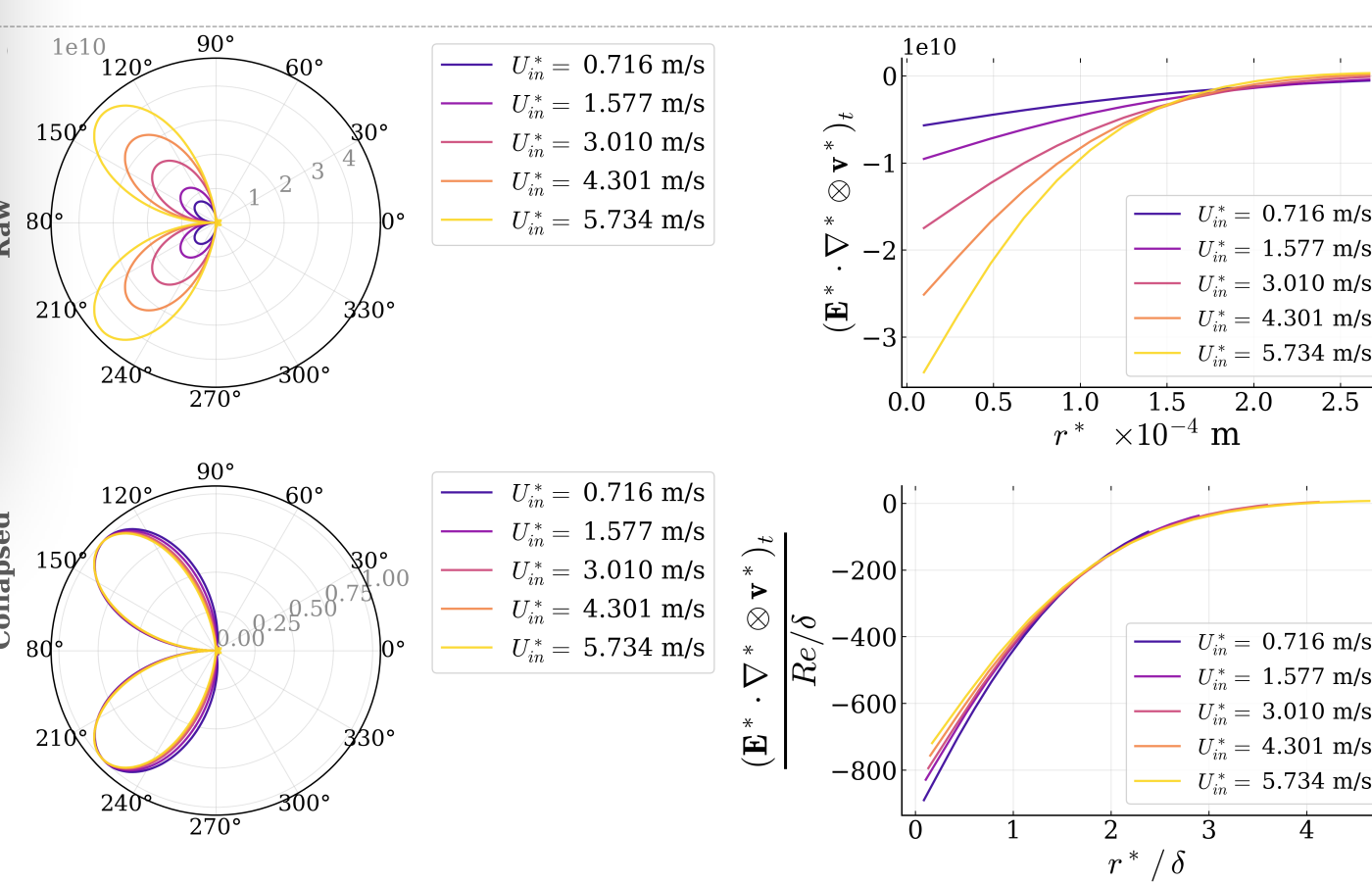
New term analysis :

$$\frac{\epsilon_0}{\mu} \int_{\Omega} \mathbf{E} \cdot \nabla \mathbf{v}$$

- Globally **detrimental** :
- Propulsive at the emitter
- Detrimental at the collector

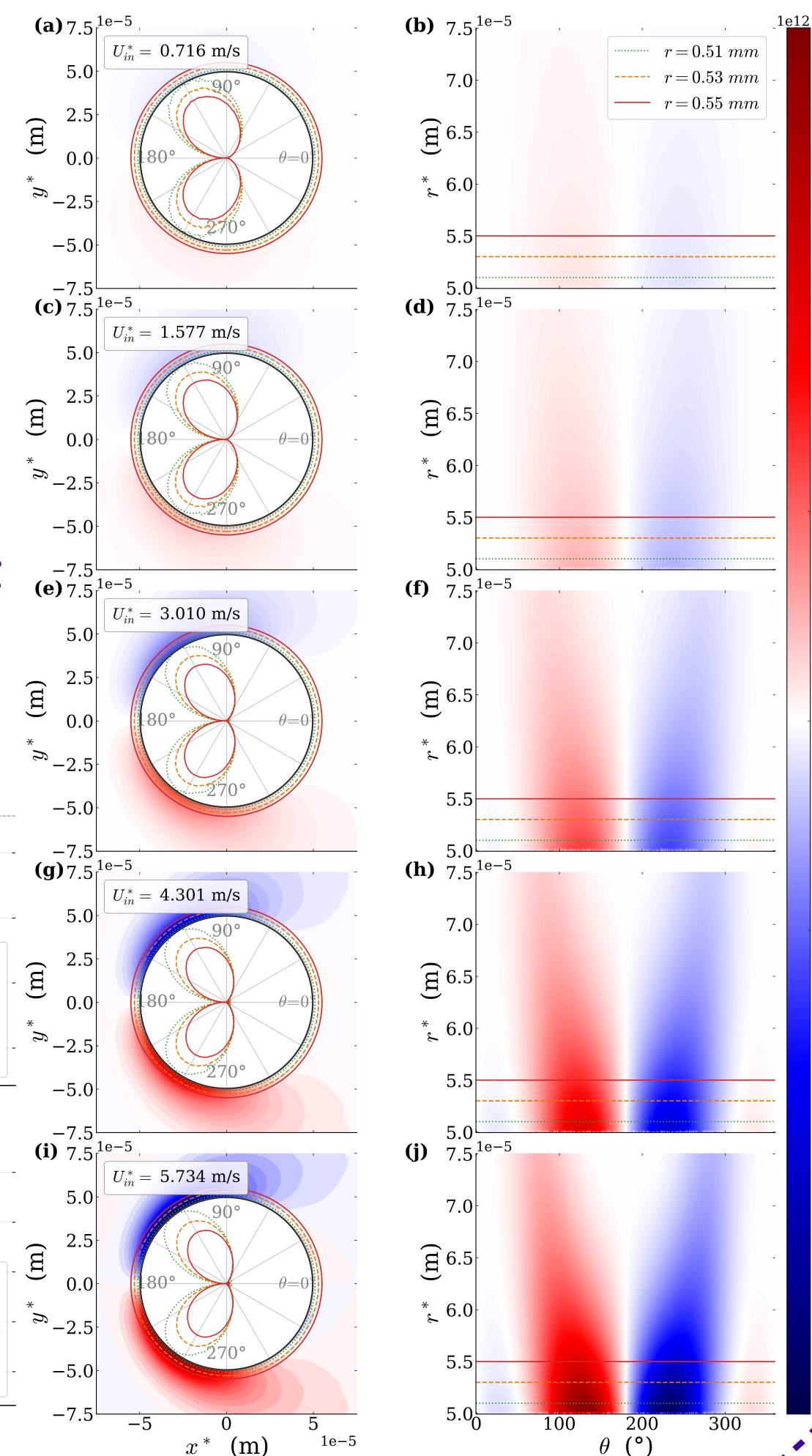
- Dominated by the **boundary layers** of the flow at the electrodes : scales as $R_e^{3/2}$

Azimuthal and Radial cut at collector, resp. :



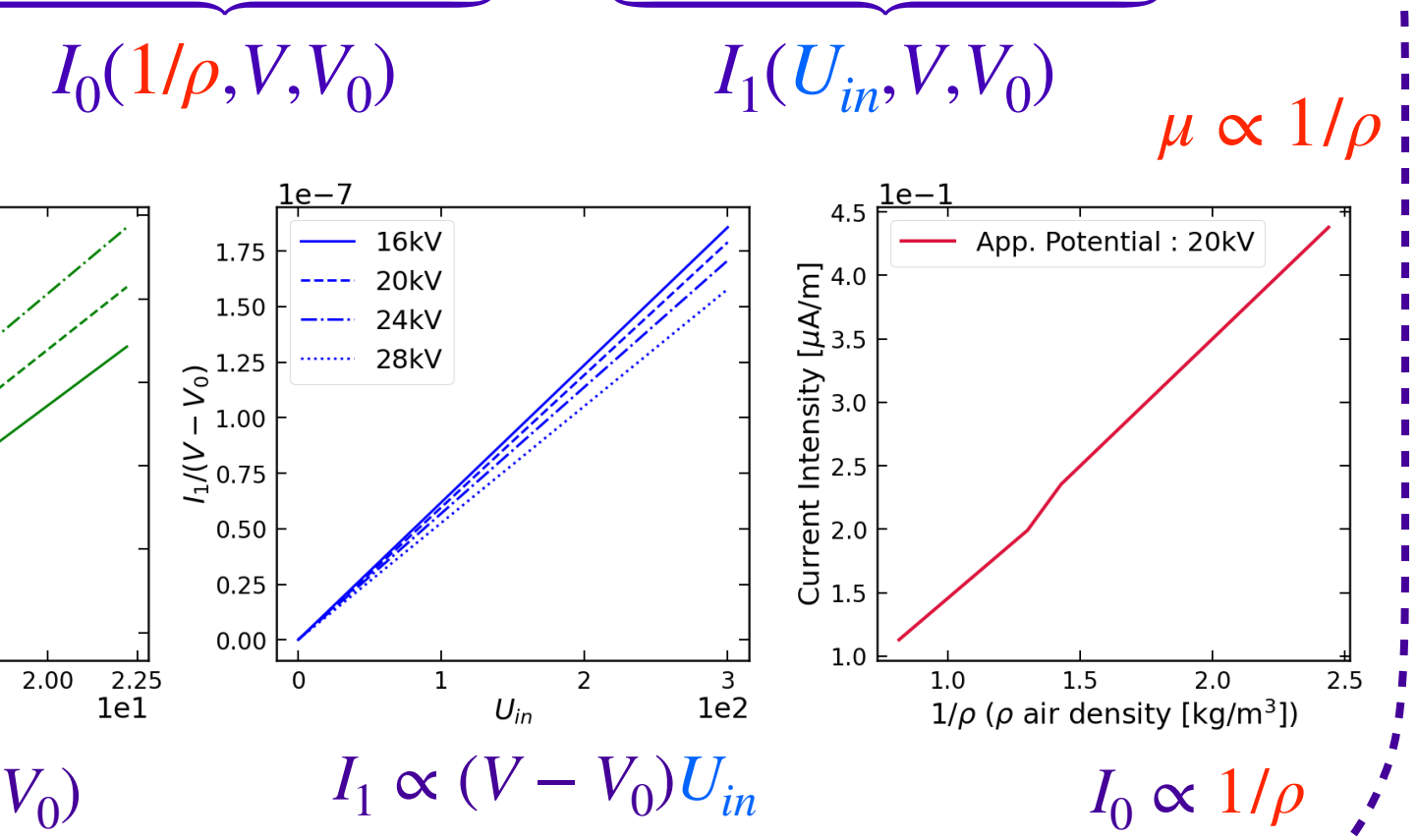
EHD Thrust

$\mathbf{E} \cdot \nabla \mathbf{v}$ field near the emitter :

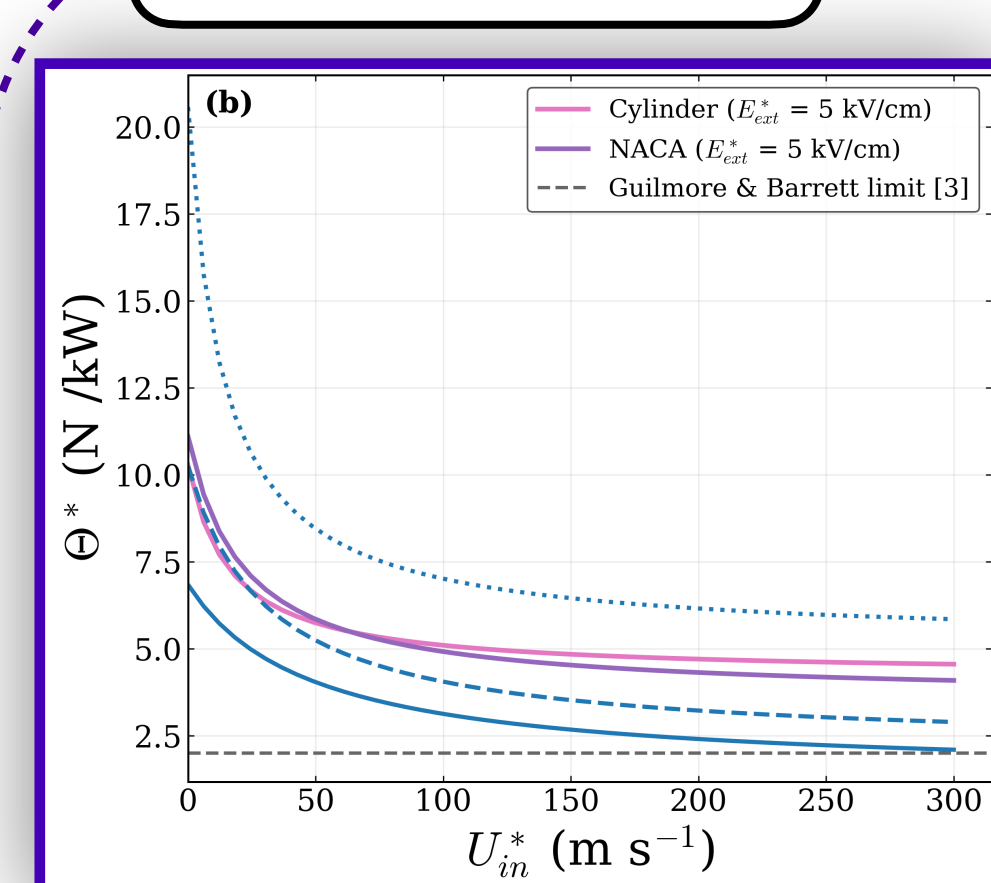


Verification of Chapman-Townsend Law

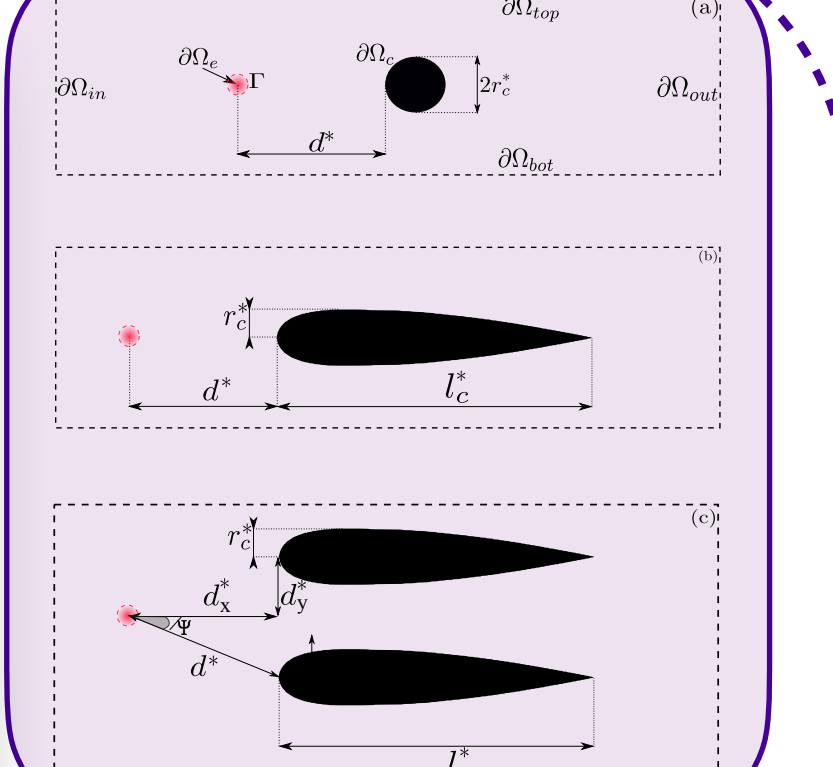
$$I = K\mu V(V - V_0) + CU_{in}(V - V_0)$$



Thrust-to-Power



Configurations



- Always **above the 2 N/kW** limit for propulsion applications

Kinetic Transfer Efficiency

$$\eta_K = \frac{\int_{\Omega} \rho \mathbf{E} \cdot \mathbf{u}}{\int_{\Omega} \rho \mathbf{E} \cdot \mathbf{u} + \int_{\Omega} \rho \mathbf{E} \cdot \mu \mathbf{E}} = \frac{\mathcal{P}_E}{\mathcal{P}_E + \mathcal{P}_T} \Leftrightarrow \frac{1}{\eta_K} = 1 + \frac{\mathcal{P}_T}{\mathcal{P}_E}$$

Linear perturbative expansion over M_ρ : $\frac{1}{\eta_K} \approx 1 + \frac{\mathcal{P}_K^{0,0} + M_a \mathcal{P}_K^{0,1} + M_\rho \mathcal{P}_K^{1,0}}{\mathcal{P}_E^{0,0} + M_\rho \mathcal{P}_E^{1,0}}$

Retrieves linear form :

$$\frac{\eta_K^\infty}{\eta_K} = 1 + \eta_K^\infty K \frac{U_e^*}{U_{in}^*}$$

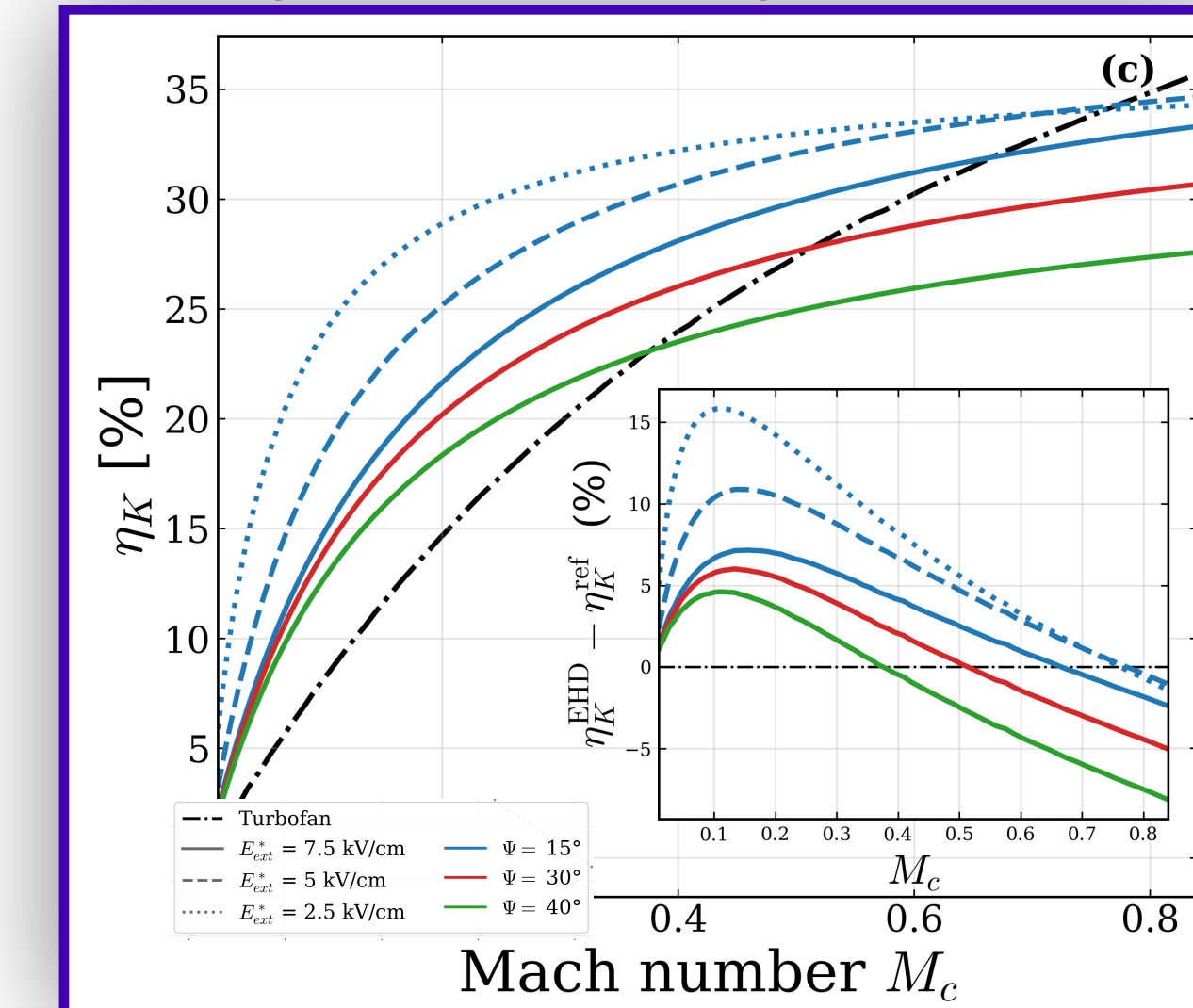
$$\text{And } \eta_K^\infty = \left(1 + \frac{\mathcal{P}_K^{1,0}}{\mathcal{P}_K^{0,0}} - \frac{\mathcal{P}_K^{0,1}}{(\mathcal{P}_K^{0,0})^2}\right)^2$$

Thus :

$$\eta_K = \frac{\eta_K^\infty}{1 + \eta_K^\infty K \frac{U_e^*}{U_{in}^*}}$$

Comparison to Turbofan :

- 2.5 times more efficient at 0.1 Mach !
- Globally more efficient up until 0.5 Mach



$$\Theta = \frac{F_{EHD}}{IV} = \frac{Id/\mu + \int_{\Omega} \mathbf{E} \cdot \nabla \mathbf{v}}{IV}$$

Replacing with Chapman-Townsend's law :

$$\Theta = \Theta_0 + \frac{1}{\mu V} \left[\frac{k_{E \cdot \nabla \mathbf{v}}}{K(V - V_0) + CV(V - V_0)/U_{in}} \right]$$

$$\rightarrow \Theta_0 + \frac{1}{\mu V} \frac{k_{E \cdot \nabla \mathbf{v}}}{K(V - V_0)} =: \Theta_\infty$$

At $U_{in} = 0$ m/s

